Discrete Structure (CST-002)

Complete unit-wise notes - easy English, matched to the UTU syllabus.

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Unit 1 - Set Theory, Relations & Functions

What this unit covers: sets and their operations, relations, and functions - the language the rest of the subject is written in.

Key ideas: A **set** is an unordered collection with no repeats; know union, intersection, difference, complement, and the power set (2^n subsets for n elements). A **relation** from A to B is a subset of A x B. Check every relation for three properties: **reflexive** (every element relates to itself), **symmetric** (aRb implies bRa), **transitive** (aRb and bRc imply aRc). All three together = **equivalence relation** (it splits the set into disjoint classes). Reflexive + antisymmetric + transitive = **partial order**. A **function** maps every input to exactly one output: **injective** (one-one), **surjective** (onto), **bijective** (both, and only then an inverse exists).

Exam pointers: practice proving reflexive/symmetric/transitive with one clean example each; composition of functions (f o g) and inverse of a bijection are repeated questions. Draw the arrow diagram before you claim injective or surjective.

Unit 2 - Propositional Calculus & Lattices

What this unit covers: propositional logic (truth tables, equivalences, normal forms) and lattices - posets where every pair has a meet and a join.

Key ideas: A **proposition** is a statement that is true or false. Learn the connectives: AND, OR, NOT, implication ($p \rightarrow q$ is false only when p is true and q is false), biconditional. A formula true in every row is a **tautology**, false in every row a **contradiction**. Two formulas are **logically equivalent** when their truth tables match - memorise De Morgan's laws, distributive laws, and $p \rightarrow q \equiv \neg p \vee q$. **DNF** = OR of ANDs, **CNF** = AND of ORs. A **lattice** is a poset where every pair has a least upper bound (join) and greatest lower bound (meet). Then: **bounded** (has 0 and 1), **distributive**, **complemented** (every element has a complement) - all three together = **Boolean algebra**.

Exam pointers: truth-table questions are free marks - be systematic. For lattices, practice checking "is this poset a lattice?" by finding joins/meets of every pair from the Hasse diagram, and identifying sublattices and special lattices.

Extra note - Special lattices - 1-minute notes:

Lattice = a poset in which every pair of elements has a **meet (greatest common lower bound)** and a **join (least common upper bound)**.

- **Bounded lattice:** has a greatest element (1) and a least element (0). Example: in the power set P(S), S is the greatest and the empty set is the least.
- **Distributive lattice:** a AND (b OR c) = (a AND b) OR (a AND c) - the operation distributes both ways. The power set with union/intersection is distributive.
- **Complemented lattice:** every element has a complement (x OR x' = 1, x AND x' = 0). In a power set, the complement is the set difference.

Exam tip: from the Hasse diagram, check every pair for a meet and a join. If even one pair has no meet or no join, it is NOT a lattice.

Unit 3 - Counting & Recurrence Relations

What this unit covers: counting techniques and recurrence relations - how to count without listing, and how to solve sequences defined by their previous terms.

Key ideas: Product rule (choices multiply for "and"), **sum rule** (choices add for exclusive "or"). **Permutation** P(n,r) = n!/(n-r)! when order matters; **combination** C(n,r) = n!/(r!(n-r)!) when it does not. **Pigeonhole:** n+1 pigeons in n holes force one hole with at least two - great for short proof questions. **Inclusion-exclusion:** $|A \cup B| = |A| + |B| - |A \cap B|$. For **recurrence relations** like $a_n = 5a_{n-1} - 6a_{n-2}$: write the characteristic equation, find roots r_1, r_2; distinct roots give $a_n = A \cdot r_1^n + B \cdot r_2^n$, repeated root r gives $a_n = (A + Bn)r^n$; fix A, B from the initial conditions.

Exam pointers: solving a linear homogeneous recurrence with given initial conditions is the single most repeated numerical - practise 4-5 until the method is automatic. Know the difference between "arrangement" (permutation) and "selection" (combination) word problems.

Unit 4 - Algebraic Structures (Groups & Rings)

What this unit covers: algebraic structures - semigroups, monoids, groups, subgroups, cyclic groups, cosets and Lagrange's theorem, then rings, integral domains and fields.

Key ideas: A set with an operation is a **semigroup** if the operation is closed and associative; add an **identity** element and it is a **monoid**; add an **inverse** for every element and it is a **group**. If the operation

also commutes, it is **abelian**. A **subgroup** is a subset that is itself a group (check closure and inverses). A group generated by one element is **cyclic**. **Lagrange's theorem:** the order of a subgroup divides the order of the group - used constantly. Moving up: a **ring** has two operations (like + and ×) where + is an abelian group and × distributes over +; an **integral domain** adds commutativity, identity and no zero-divisors; a **field** is a ring where every non-zero element has a multiplicative inverse.

Exam pointers: "prove (G, *) is a group" = tick the four axioms one by one with named steps. Cosets + Lagrange and the ring/domain/field chain are favourite theory questions - see the simple notes written for Integral Domain & Field below in this page.

Extra note - Integral Domain & Field - made simple:

First, a ring: a set of numbers where **both + and x** work nicely (like integers: 1+2=3, 2x3=6).

Zero divisor: two non-zero numbers whose product is **0**. This never happens with normal integers. But in Z6 (clock numbers 0-5): `2 x 3 = 6 = 0` - so 2 and 3 are zero divisors.

Integral domain: a ring with **no** zero divisors. Example: Z (all integers) - here a x b = 0 only when a or b is itself 0.

Field: a ring where every non-zero number has its **inverse** in the same set - meaning you can divide.

> *Small example: in Z5 (numbers 0-4), check: 2 x 3 = 6 = 1, so the inverse of 2 is 3. And 4 x 4 = 16 = 1, so 4 is its own inverse. Every element has an inverse - **Z5 is a field**. Z (integers) is not a field, because the inverse of 2 is 1/2, which is not an integer.*

Rule to remember: every field is an integral domain (if inverses exist, zero divisors cannot). Every finite integral domain is always a field.

Unit 5 - Graphs & Trees

What this unit covers: graph theory and trees - vertices, edges, paths, Euler and Hamilton circuits, planarity, spanning trees and tree traversals.

Key ideas: A **graph** G = (V, E). The **handshaking lemma**: sum of all degrees = 2 x number of edges (so the number of odd-degree vertices is always even). An **Euler circuit** uses every edge exactly once and exists when every vertex has even degree; a **Hamilton circuit** visits every vertex exactly once (no easy test - remember that contrast). A graph is **bipartite** if vertices split into two sets with no edge inside a set (equivalently, no odd cycle). **Planar** graphs draw without crossings; Euler's formula: v - e + r = 2. A **tree** is a connected graph with no cycles: n vertices always give n - 1 edges. A **spanning tree** touches every vertex; **Kruskal** (cheapest edge first, skip cycles) and **Prim** (grow one tree from a start vertex) find the minimum spanning tree. Binary tree traversals: **preorder** (root-left-right), **inorder** (left-root-right), **postorder** (left-right-root).

Exam pointers: Euler/Hamilton existence checks, Euler's formula numerics, one Kruskal/Prim step-by-step MST, and one traversal question appear almost every year.

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